

MECHANICAL PROPERTIES OF SOLIDS

Important formulas and concepts directly from NCERT–

Elasticity = if body regain its original shape and size , when applied force is removed ,this property is elasticity

Plasticity = if a force apply on a body, and they have no gross tendency to regain their Previous shape ,and they get permanently deformed , this property is called plasticity e.g.putty and mud are ideal plastics

1] stress = restoring force/area = $\frac{F}{A}$

2] 3 types of stress –

1] longitudinal stress = if force applied **normal** to cross sectional area then force applied per unit area is longitudinal stress

❖ It is of 2 types-

1] tensile stress – if cylinder is stretched by 2 equal forces applied normal to its cross sectional area

2] compressive stress - if cylinder is compressed by action of force normal to its area

2] shearing or tangential stress = if 2 equal and opposite forces applied **parallel** to Its cross sectional area then restoring force per unit area

3] hydraulic stress = if a body is compressed or stretched uniformly on all sides Then force applied acts in perpendicular direction at each point of surface,then body Body develops internal restoring force ,this force per unit area is hydraulic stress [no Change in shape]

3] 3 types of strain –

1] longitudinal strain = $\frac{\text{change in length}}{\text{Original length}} = \frac{\Delta L}{L}$

Original length

2] shearing strain = $\frac{\text{relative displacement of the faces}}{\text{Length of cylinder}} = \frac{\Delta x}{L} = \tan \theta \approx \theta$

Length of cylinder

3] volumetric strain = $\frac{\text{change in volume}}{\text{Original volume}} = \frac{\Delta V}{V}$

Original volume

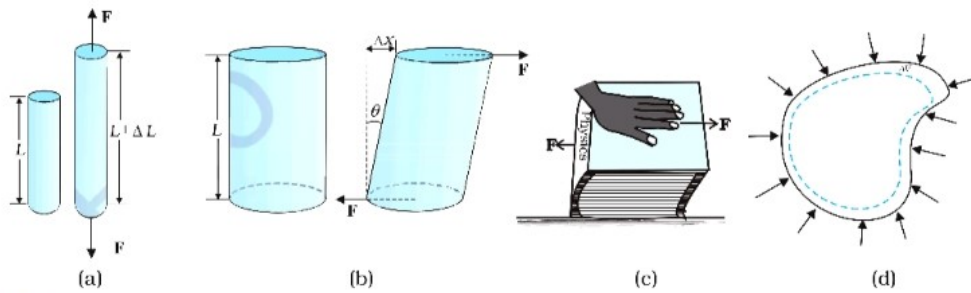


Fig. 8.1 (a) A cylindrical body under tensile stress elongates by ΔL (b) Shearing stress on a cylinder deforming it by an angle θ (c) A body subjected to shearing stress (d) A solid body under a stress normal to the surface at every point (hydraulic stress). The volumetric strain is $\Delta V/V$, but there is no change in shape.

❖ Hooke's law = stress $\propto$ strain

Valid for most materials but there are some materials which do not exhibit this linear relationship

❖ Stress strain curve =

- Vary from material to material
- Substances like tissue of aorta, rubber etc. which can be stretched to cause large strains, are called elastomers.
- In OA region, hooke's law is obeyed and solid behave as elastic body.
- Yield point also known as **elastic limit** and the corresponding strength is known as **yield strength**.
- If load increased further after elastic limit, the stress exceeds the yield strength and strain increase rapidly even for a small change in stress, BD region show this.
- B/w B and D, body doesn't regain its original dimension, in this case even stress zero, strain is not zero. material is said to have **permanent set**. the deformation is PLASTIC.
- Point D is **ultimate tensile strength**, beyond this point, additional strain is developed even by a reduced force and fracture occurs at E.

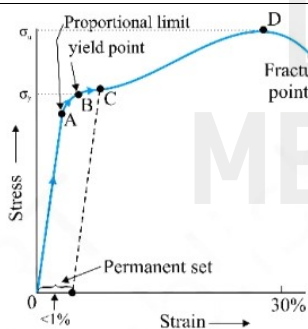


Fig. 8.2 A typical stress-strain curve for a metal.

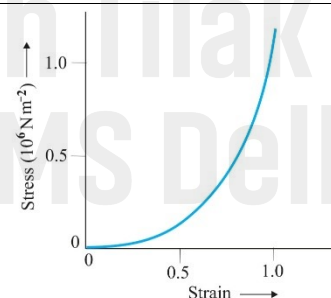


Fig. 8.3 Stress-strain curve for the elastic tissue of Aorta, the large tube (vessel) carrying blood from the heart.

- The ratio of stress and strain is called **modulus of elasticity** [characteristic of a material]
- The magnitude of strain produced is same **while stress is tensile or compressive**

❖ **Young's modulus** = ratio of tensile stress to longitudinal strain

$$Y = \sigma / \epsilon$$

$$Y = [F/A] / [\Delta L/L] = [F \times L] / [A \times \Delta L]$$

Unit of Y=same as pressure as strain is unitless

➤ Steel is preferred in heavy duty machines and in structural designs as it is more elastic than aluminium, copper and brass

➤ Wood, bone, concrete and glass have rather small young's modulus

$$Y \text{ of the material of a wire} = Mg/\pi r^2 \cdot L/\Delta L$$

➤ Y depend upon-

1] impurity 2] temperature 3] nature of crystal solid

➤ Y doesn't defined for liquid or gas [only for solids]

➤ Y is inversely prop. To temperature

❖ **Shear modulus [modulus of rigidity]**= ratio of shearing stress to shearing strain

$$G = [F/A] / [\Delta x/L] = [F \times L] / [A \times \Delta x]$$

Or

$$G = [F/A] / \theta$$

Shearing stress, $\sigma_s = G \times \theta$

➤ Shear modulus is generally less than young's modulus [$G = Y/3$]

➤ Also not defined for liquid or gas

❖ **Bulk modulus** = the ratio of hydraulic stress to volumetric strain

$$B = -p / [\Delta V/V]$$

➤ Neg. sign indicates that with increase in pressure, a decrease in volume occurs

➤ For a system in equilibrium, B is always positive

Compressibility[k] = reciprocal of B

$$K = 1/B$$

➤ B for solids are much more than the bulk modulus for gases

➤ Thus solids are less compressible and gases are most compressible [gases are about one million times more compressible than solids]

➤ Gases have large compressibility which vary with pressure and temp

➤ the incompressibility of solids is primarily due to tight coupling b/w neighbouring atoms

Table 8.4 Stress, strain and various elastic moduli

Type of stress	Stress	Strain	Change in		Elastic Modulus	Name of Modulus	State of Matter
			shape	volume			
Tensile or compressive ($\sigma = F/A$)	Two equal and opposite forces perpendicular to opposite faces	Elongation or compression parallel to force direction ($\Delta L/L$) (longitudinal strain)	Yes	No	$Y = (F L) / (A \Delta L)$	Young's modulus	Solid
Shearing ($\sigma_s = F/A$)	Two equal and opposite forces parallel to opposite surfaces forces in each case such that total force and total torque on the body vanishes	Pure shear, θ	Yes	No	$G = F/(A \theta)$	Shear modulus or modulus of rigidity	Solid
Hydraulic	Forces perpendicular everywhere to the surface, force per unit area (pressure) same everywhere.	Volume change (compression or elongation) ($\Delta V/V$)	No	Yes	$B = -p/(\Delta V/V)$	Bulk modulus	Solid, liquid and gas

❖ Poisson's ratio =

- The strain perpendicular to the applied force is called lateral strain
- Given by simon poisson
- Within elastic limit, lateral strain is directly prop. to longitudinal strain
- The ratio of lateral strain to longitudinal strain in a stretched wire is known as poisson ratio
- Lateral strain = $\frac{\text{contraction of diameter under stress}}{\text{Original diameter}} = \frac{\Delta d}{d}$
- Poisson's ratio = $[\Delta d / d] / [\Delta L / L]$
- It is a pure no. and has no dimensions or units
- Its value **depends only on nature of material**
- For steel, value is b/w 0.28 and 0.3
For aluminium , value is 0.33
- By hooke's law, $F/A = Y \cdot \Delta L/L$
 $F/A = Y \cdot \frac{\Delta d}{d}$
Poisson ratio

❖ Elastic Potential Energy in stretched wire = when a wire is put under a tensile stress , work is done against the interatomic forces. This work is stored in the wire in the form of elastic potential energy

$$\begin{aligned}
 \text{Elastic potential energy} &= \frac{1}{2} \times \text{force} \times \text{elongation} \\
 &= \frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume of wire} \\
 &= \frac{1}{2} \times \text{strain}^2 \times \text{young's modulus} \\
 &\quad \times \text{volume of wire}
 \end{aligned}$$

$$\begin{aligned}
 \text{Elastic potential energy per unit volume} &= \frac{1}{2} \times \text{stress} \times \text{strain} \\
 &= \frac{1}{2} \times \text{strain}^2 \times Y \\
 &= \frac{1}{2} \times \text{stress}^2 / Y
 \end{aligned}$$

SUMMARY

1. Stress is the restoring force per unit area and strain is the fractional change in dimension. In general there are three types of stresses (a) tensile stress — longitudinal stress (associated with stretching) or compressive stress (associated with compression), (b) shearing stress, and (c) hydraulic stress.
2. For small deformations, stress is directly proportional to the strain for many materials. This is known as Hooke's law. The constant of proportionality is called modulus of elasticity. Three elastic moduli viz., Young's modulus, shear modulus and bulk modulus are used to describe the elastic behaviour of objects as they respond to deforming forces that act on them.
A class of solids called elastomers does not obey Hooke's law.
3. When an object is under tension or compression, the Hooke's law takes the form
$$F/A = Y\Delta L/L$$
where $\Delta L/L$ is the tensile or compressive strain of the object, F is the magnitude of the applied force causing the strain, A is the cross-sectional area over which F is applied (perpendicular to A) and Y is the Young's modulus for the object. The stress is F/A .
4. A pair of forces when applied parallel to the upper and lower faces, the solid deforms so that the upper face moves sideways with respect to the lower. The horizontal displacement ΔL of the upper face is perpendicular to the vertical height L . This type of deformation is called shear and the corresponding stress is the shearing stress. This type of stress is possible only in solids.
In this kind of deformation the Hooke's law takes the form
$$F/A = G \times \Delta L/L$$
where ΔL is the displacement of one end of object in the direction of the applied force F , and G is the shear modulus.
5. When an object undergoes hydraulic compression due to a stress exerted by a surrounding fluid, the Hooke's law takes the form
$$p = B (\Delta V/V),$$
where p is the pressure (hydraulic stress) on the object due to the fluid, $\Delta V/V$ (the volume strain) is the absolute fractional change in the object's volume due to that pressure and B is the bulk modulus of the object.

POINTS TO PONDER

1. In the case of a wire, suspended from ceiling and stretched under the action of a weight (F) suspended from its other end, the force exerted by the ceiling on it is equal and opposite to the weight. However, the tension at any cross-section A of the wire is just F and not $2F$. Hence, tensile stress which is equal to the tension per unit area is equal to F/A .
2. Hooke's law is valid only in the linear part of stress-strain curve.
3. The Young's modulus and shear modulus are relevant only for solids since only solids have lengths and shapes.
4. Bulk modulus is relevant for solids, liquid and gases. It refers to the change in volume when every part of the body is under the uniform stress so that the shape of the body remains unchanged.
5. Metals have larger values of Young's modulus than alloys and elastomers. A material with large value of Young's modulus requires a large force to produce small changes in its length.
6. In daily life, we feel that a material which stretches more is more elastic, but it is a misnomer. In fact material which stretches to a lesser extent for a given load is considered to be more elastic.
7. In general, a deforming force in one direction can produce strains in other directions also. The proportionality between stress and strain in such situations cannot be described by just one elastic constant. For example, for a wire under longitudinal strain, the lateral dimensions (radius of cross section) will undergo a small change, which is described by another elastic constant of the material (called *Poisson ratio*).
8. Stress is not a vector quantity since, unlike a force, the stress cannot be assigned a specific direction. Force acting on the portion of a body on a specified side of a section has a definite direction.

Important points from NCERT summary and points to ponder

- Young's modulus, shear modulus and bulk modulus are used to describe elastic behavior as they respond to deforming forces that act on them
- Elastomers doesn't obey hooke's ,law.
- In case of a wire suspended from ceiling and stretched under action of weight[F],then tension at any cross section of wire is just F not 2F. 2023
- Hooke's law is valid only in linear part of stress-strain curve.
- Y and G are relevant only for solids since only solids have lengths and shapes
- B is relevant for solids,liquids and gases.
- Metals have larger value of Y than alloys and elastomers. A material with large value of Y requires large force to produce small change in its length.
- In fact, the material which stretches to a lesser extent for a given load is considered to be more elastic.
- Stress is not a vector quantity , since stress can't be assigned a specific direction.

Importants from the NEET Previous Year Questions -

[1] the material whose ultimate strength point and fracture point are close then material is brittle or if far then material is ductile.

[2]the stretching of a spring is determined by the shear modulus of material of spring.

$$\begin{aligned} [3] p &= B \cdot \Delta V/V \\ Pgh &= B \cdot \Delta V/V \end{aligned}$$